

Quark-mass table + the up/down ladders + the gen-1 inversion (the m_u/m_d keystone)

Grace | 2026-07-19 | my item supporting the m_u/m_d derivation (K759). Full quark-mass table, both ladders, all ratios with BST forms, and the derived-vs-candidate classification. Discipline held (Cal's round): m_u/m_d 's loose range (0.38–0.58) admits FOUR clean forms — it MUST be derived from the doublet geometry, not matched. The gen-1 inversion is the keystone puzzle for Lyra.

The table (current masses, MSbar PDG, MeV)

	up-type	down-type
gen 1	$m_u = 2.16$	$m_d = 4.67$
gen 2	$m_c = 1270$	$m_s = 93.4$
gen 3	$m_t = 172690$	$m_b = 4180$

Down-type ladder — ANCHORED (one derived rung locks it)

ratio	value	BST form	tier
m_s/m_d	20.00	$\text{rank}^2 \cdot n_C = 20$	DERIVED (F506, banked)
m_b/m_s	44.75	$\sim?$	candidate
m_b/m_d	895	$\sim?$	candidate

The down ladder is anchored: $m_s/m_d = 20$ is forced, so m_s (and via m_b/m_s , m_b) are locked to m_d . That's why the down-type is solid.

Up-type ladder — FLOATS (no anchor yet)

ratio	value	BST form	tier
m_c/m_u	588	$\sim?$	candidate
m_t/m_c	135.98	$\sim N_{\text{max}} - 1 = 136$ (0.0%)	candidate (near-match, don't over-read)
m_t/m_b	41.31	$C_2 \cdot g = 42$	LATTICE (identified)

The up ladder has no derived anchor to m_d — which is exactly why m_u floats. Lock m_u/m_d and the whole up-type ladder (u:c:t) anchors.

★ The gen-1 inversion (the keystone puzzle — Lyra to DERIVE)

generation	up/down ratio	status
gen 1	$m_u/m_d = 0.463$	INVERTED (down heavier than up)
gen 2	$m_c/m_s = 13.60$	normal (up heavier)
gen 3	$m_t/m_b = 41.31$	normal (up heavier)

Generation 1 inverts (down > up); generations 2 and 3 don't. This is the heart of the m_u/m_d derivation — the doublet geometry must *explain why gen-1 flips*. Not a form that fits; a mechanism that inverts.

☐ $m_u/m_d = 0.463$ — the coincidence trap (Cal’s discriminator this round)

The observed range (0.38–0.58) is loose enough that four clean BST forms all fit: | form | value | dev | in range? |
 |—|—|—|—| | $N_c/g = 3/7$ | 0.429 | 7.3% | ☐ | | $1/\text{rank} = 1/2$ | 0.500 | 8.1% | ☐ | | $\text{rank}/n_C = 2/5$ | 0.400 | 13.5% | ☐ |
 | | $\sqrt{(3/14)} = \sqrt{(N_c/(\text{rank} \cdot g))}$ | 0.463 | 0.1% | ☐ |

This is the trap we killed all week (3/13, $D=10$, $\Lambda \approx m_p/N_c$). Four forms fit → fitting is not deriving. Even $\sqrt{(3/14)}$ ’s 0.1% is *suspicious*, not reassuring: a 0.1% match to a quantity with a ~40% experimental range is over-fit territory, not precision. m_u/m_d must FALL OUT of the up-down doublet geometry (the $\mathbb{H}=\text{Sp}(2)$ $T_3=\pm 1/2$ radial-moment split) with the gen-1 inversion built in — otherwise it’s matched, and Cal rejects it.

Derived-vs-candidate summary (the quark-mass sector honestly)

- DERIVED: $m_s/m_d = \text{rank}^2 \cdot n_C = 20$ (the down anchor).
- LATTICE/identified: $m_t/m_b = C_2 \cdot g = 42$.
- CANDIDATE (fit, not derived): m_u/m_d (4 forms fit — must be derived), $m_c/m_u=588$, $m_b/m_s=44.75$, $m_t/m_c \approx 136$.
- The keystone: derive m_u/m_d + the gen-1 inversion → m_u solid → up-type ladder anchors → quark-mass ratios (the Statement-B counterexample) start to fall.

— Grace, 2026-07-19. Quark-mass table + ladders: down-type anchored ($m_s/m_d=20$ derived), up-type floats (no derived anchor → m_u soft). The keystone is m_u/m_d + the gen-1 inversion (gen-1 down>up, gens 2,3 up>down). DISCIPLINE: 4 clean forms fit the loose m_u/m_d range → it must be DERIVED from the doublet geometry (Lyra), not matched; even $\sqrt{(3/14)}$ ’s 0.1% is over-fit-suspicious. Cal’s derived-vs-matched call.